ORF 524 — Final Exam

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Contents

Notes:

- Duration of examination: 180 minutes.
  An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
  Please provide as much detail as possible in your answers.
  Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
  A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
  This is otherwise a closed-book and individual examination.

1 Best Linear Approximation (15 points)

Let Y be a scalar random variable and $\mathbf{X}$ be a random vector in $\mathbb{R}^d$. Assume that $\mathbb{E}[Y^2] < \infty$ and $g \in \mathcal{G} = \{g : \mathbb{E}[g(\mathbf{X})^2] < \infty\}$. We are interested in the predictor $g^* \in \mathcal{G}$ that minimizes the Mean Squared Error (MSE), defined as the risk functional $R(g) = \mathbb{E}[(Y - g(\mathbf{X}))^2]$. Let $\mu(\mathbf{X}) = \mathbb{E}[Y|\mathbf{X}]$.

1. *General Decomposition.* Show that $R(g) = \mathbb{E}[\mathbb{V}ar[Y|\mathbf{X}]] + \mathbb{E}[(\mu(\mathbf{X}) - g(\mathbf{X}))^2]$ for all $g \in \mathcal{G}$.

   Solution: Follows by expanding the square $R(g) = \mathbb{E}[(Y - \mu(\mathbf{X}) + \mu(\mathbf{X}) - g(\mathbf{X}))^2]$, and noting that $\mathbb{E}[(Y - \mu(\mathbf{X}))(\mu(\mathbf{X}) - g(\mathbf{X}))] = 0$ and $\mathbb{E}[(Y - \mu(\mathbf{X}))^2] = \mathbb{E}[\mathbb{V}ar[Y|\mathbf{X}]]$ by the law of iterated expectations.

2. *Linear Model.* Assume that $\mu(\mathbf{X}) = \mathbf{X}^\top \boldsymbol{\beta}_0$ for some unique parameter vector $\boldsymbol{\beta}_0 \in \mathbb{R}^d$, and consider all candidates $g \in \mathcal{G}_{\mathrm{L}} = \{g : g(\mathbf{X}) = \mathbf{X}^\top \boldsymbol{\beta}, \boldsymbol{\beta} \in \mathbb{R}^d\}$. Give sufficient conditions to ensure that $\mathcal{G}_{\mathrm{L}} \subseteq \mathcal{G}$, and show that $R(g)$ is determined by a quadratic form involving the second moment matrix of the covariates, $\boldsymbol{\Sigma}_{\mathbf{X}} = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$. Give sufficient conditions to ensure that μ is the unique solution of $\min_{g \in \mathcal{G}_{\mathrm{L}}} R(g)$.

   Solution: Because $\mathbb{E}[g(\mathbf{X})^2] = \mathbb{E}[(\mathbf{X}^\top \boldsymbol{\beta})^2] = \boldsymbol{\beta}^\top \mathbb{E}[\mathbf{X}\mathbf{X}^\top]\boldsymbol{\beta} < \|\boldsymbol{\beta}\| \|\boldsymbol{\Sigma}_{\mathbf{X}}\|$, the condition $\mathbb{E}[\|\mathbf{X}\|^2] < \infty$ guarantees that $g \in \mathcal{G}$, where $\|\cdot\|$ denotes the Euclidean norm. Next,

   $$\begin{aligned}\mathbb{E}[(\mu(\mathbf{X}) - g(\mathbf{X}))^2] &= \mathbb{E}[(\mathbf{X}^\top \boldsymbol{\beta}_0 - \mathbf{X}^\top \boldsymbol{\beta})^2] = \mathbb{E}[(\boldsymbol{\beta}_0 - \boldsymbol{\beta})^\top \mathbf{X}\mathbf{X}^\top (\boldsymbol{\beta}_0 - \boldsymbol{\beta})] \\ &= (\boldsymbol{\beta} - \boldsymbol{\beta}_0)^\top \boldsymbol{\Sigma}_{\mathbf{X}} (\boldsymbol{\beta} - \boldsymbol{\beta}_0).\end{aligned}$$

   Since the first term in the general decomposition does not depend on $\boldsymbol{\beta}$ (irreducible error), minimizing the MSE is equivalent to minimizing this quadratic form. If $\boldsymbol{\Sigma}_{\mathbf{X}}$ is positive definite, this quadratic form is strictly positive for all $\boldsymbol{\beta} \neq \boldsymbol{\beta}_0$, and zero if and only if $\boldsymbol{\beta} = \boldsymbol{\beta}_0$. Thus, $\boldsymbol{\beta}_0$ is the unique minimizer under the (sufficient) condition that $\boldsymbol{\Sigma}_{\mathbf{X}}$ is positive definite.

3. *Orthogonality Condition and Interpretation.* Consider now a general conditional mean $\mu(\mathbf{X})$, not necessarily equal to $\mathbf{X}^\top \boldsymbol{\beta}_0$. What is the interpretation for the solution of $\min_{g \in \mathcal{G}_{\mathrm{L}}} R(g)$?

   Solution: Using the general decomposition,

   $$\begin{aligned}\min_{g \in \mathcal{G}_{\mathrm{L}}} R(g) &= \mathbb{E}[\mathbb{V}ar[Y|\mathbf{X}]] + \min_{g \in \mathcal{G}_{\mathrm{L}}} \mathbb{E}[(\mu(\mathbf{X}) - g(\mathbf{X}))^2] \\ &= \mathbb{E}[\mathbb{V}ar[Y|\mathbf{X}]] + \min_{\boldsymbol{\beta} \in \mathbb{R}^d} \mathbb{E}[(\mu(\mathbf{X}) - \mathbf{X}^\top \boldsymbol{\beta})^2].\end{aligned}$$

   Therefore, the solution is the best mean square approximation of $\mu(\mathbf{X})$ by a linear function of $\mathbf{X}$, given by $g^*(\mathbf{X}) = \mathbf{X}^\top \boldsymbol{\beta}^*$ with $\boldsymbol{\beta}^* \in \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^d} \mathbb{E}[(\mu(\mathbf{X}) - \mathbf{X}^\top \boldsymbol{\beta})^2]$. In particular, if $\mu(\mathbf{X}) = \mathbf{X}^\top \boldsymbol{\beta}_0$, then $\boldsymbol{\beta}^* = \boldsymbol{\beta}_0$.

2 Quasi-Maximum Likelihood and Poisson Regression (35 points)

Let $(Y_i, \mathbf{X}_i^\top)$ for $i = 1, \ldots, n$ be independent and identically distributed observations, where $Y_i \in \mathbb{N}_0$ is a count response variable and $\mathbf{X}_i \in \mathbb{R}^p$ is a vector of covariates. Assume the conditional mean is correctly specified as:

$$\mathbb{E}[Y_i|\mathbf{X}_i] = \exp(\mathbf{X}_i^\top \boldsymbol{\beta}_0) \equiv \mu_i(\boldsymbol{\beta}_0),$$

where $\boldsymbol{\beta}_0$ is the population parameter vector. However, the population conditional distribution of $Y_i|\mathbf{X}_i$ is **not** necessarily Poisson. The conditional variance is denoted by $\mathbb{V}ar[Y_i|\mathbf{X}_i] = \sigma^2(\mathbf{X}_i)$.

This question considers two estimators for $\boldsymbol{\beta}_0$:

- *The Poisson Quasi-Maximum Likelihood Estimator*: $\widehat{\boldsymbol{\beta}}_{\texttt{QMLE}}$, obtained by maximizing the Poisson log-likelihood despite potential misspecification of the distribution.
- *The Weighted Non-Linear Least Squares Estimator*: $\widehat{\boldsymbol{\beta}}_{\texttt{WNLS}}$, obtained by minimizing a weighted sum of non-linear squared residuals.

Let $\nabla_{\mathbf{w}} g(\mathbf{w})$ and $\nabla^2_{\mathbf{w}} g(\mathbf{w})$ denote the gradient and Hessian of the function $g : \mathbb{R}^d \mapsto \mathbb{R}$, respectively. Please answer the following questions.

1. *Poisson QMLE: Score.* Write down the Poisson log-likelihood function $\ell(\boldsymbol{\beta})$, that is, the log-likelihood under the assumption that $Y_i|\mathbf{X}_i \sim \mathsf{Poisson}(\mu_i(\boldsymbol{\beta}_0))$. Derive the score function $\mathbf{S}_n(\boldsymbol{\beta}) = \nabla_{\boldsymbol{\beta}}\ell(\boldsymbol{\beta})$, and show that $\mathbf{S}_n(\boldsymbol{\beta}_0)$ has an expectation of zero, regardless of the population conditional distribution of $Y_i|\mathbf{X}_i$.

   Solution: The Poisson log-likelihood for a single observation i is given by $\ell_i(\boldsymbol{\beta}) = Y_i\mathbf{X}_i^\top\boldsymbol{\beta} - \exp(\mathbf{X}_i^\top\boldsymbol{\beta}) - \log(Y_i!)$. The total log-likelihood is:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \Big(Y_i\mathbf{X}_i^\top\boldsymbol{\beta} - \exp(\mathbf{X}_i^\top\boldsymbol{\beta}) - \log(Y_i!)\Big).$$

   Differentiating with respect to $\boldsymbol{\beta}$:

$$\mathbf{S}_n(\boldsymbol{\beta}) = \nabla_{\boldsymbol{\beta}}\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \mathbf{X}_i\Big(Y_i - \exp(\mathbf{X}_i^\top\boldsymbol{\beta})\Big) = \sum_{i=1}^n \mathbf{X}_i(Y_i - \mu_i(\boldsymbol{\beta})).$$

   To show the expectation is zero at $\boldsymbol{\beta}_0$:

$$\mathbb{E}[\mathbf{S}_n(\boldsymbol{\beta}_0)] = \sum_{i=1}^n \mathbb{E}\Big[\mathbb{E}[\mathbf{X}_i(Y_i - \mu_i(\boldsymbol{\beta}_0))|\mathbf{X}_i]\Big] = \sum_{i=1}^n \mathbb{E}\Big[\mathbf{X}_i(\underbrace{\mathbb{E}[Y_i|\mathbf{X}_i]}_{\mu_i(\boldsymbol{\beta}_0)} - \mu_i(\boldsymbol{\beta}_0))\Big] = 0.$$

2. *Poisson QMLE: Hessian.* Derive the Hessian matrix $\mathbf{H}_n(\boldsymbol{\beta}) = \nabla^2_{\boldsymbol{\beta}}\ell(\boldsymbol{\beta})$. Let $\boldsymbol{\Omega}_0 = -\mathbb{E}[\nabla^2_{\boldsymbol{\beta}}\ell(\boldsymbol{\beta}_0)]$ and $\boldsymbol{\Sigma}_0 = \mathbb{V}ar[\mathbf{S}_n(\boldsymbol{\beta}_0)]$. Show that the information equality holds ($\boldsymbol{\Omega}_0 = \boldsymbol{\Sigma}_0$) under the assumption that $Y_i|\mathbf{X}_i \sim \mathsf{Poisson}(\mu_i(\boldsymbol{\beta}_0))$. Explain briefly why $\boldsymbol{\Omega}_0 \neq \boldsymbol{\Sigma}_0$ in the general case.

Solution: The Hessian is the derivative of the score:

$$\mathbf{H}_n(\boldsymbol{\beta}) = \nabla_{\boldsymbol{\beta}} \mathbf{S}_n(\boldsymbol{\beta}) = \sum_{i=1}^{n} -\mathbf{X}_i \mathbf{X}_i^\top \exp(\mathbf{X}_i^\top \boldsymbol{\beta}) = \sum_{i=1}^{n} -\mathbf{X}_i \mathbf{X}_i^\top \mu_i(\boldsymbol{\beta}).$$

The matrices $\boldsymbol{\Omega}_0$ and $\boldsymbol{\Sigma}_0$ are:

$$\boldsymbol{\Omega}_0 = -\mathbb{E}[\nabla^2_{\boldsymbol{\beta}} \ell_i(\boldsymbol{\beta}_0)] = \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top \mu_i(\boldsymbol{\beta}_0)],$$

and

$$\begin{aligned}\boldsymbol{\Sigma}_0 &= \mathbb{V}ar[\nabla_{\boldsymbol{\beta}} \ell_i(\boldsymbol{\beta}_0)] = \mathbb{E}[\nabla_{\boldsymbol{\beta}} \ell_i(\boldsymbol{\beta}_0) \nabla_{\boldsymbol{\beta}} \ell_i(\boldsymbol{\beta}_0)^\top] \\ &= \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top (Y_i - \mu_i(\boldsymbol{\beta}_0))^2] = \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top \mathbb{V}ar(Y_i | \mathbf{X}_i)].\end{aligned}$$

If the Poisson assumption holds, $\mathbb{V}ar(Y_i|\mathbf{X}_i) = \mu_i(\boldsymbol{\beta}_0)$. Substituting this into $\boldsymbol{\Sigma}_0$ gives $\boldsymbol{\Sigma}_0 = \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^\top \mu_i(\boldsymbol{\beta}_0)] = \boldsymbol{\Omega}_0$. Thus, the information equality holds.

If there is overdispersion, $\mathbb{V}ar(Y_i|\mathbf{X}_i) > \mu_i(\boldsymbol{\beta}_0)$, which implies $\boldsymbol{\Sigma}_0 > \boldsymbol{\Omega}_0$ (in the positive semidefinite sense). The standard errors derived from $\boldsymbol{\Omega}_0^{-1}$ alone will be too small.

3. *Poisson QMLE: Asymptotic Normality.* Using standard M-estimation theory (e.g., Taylor expansion of the score function), state the asymptotic distribution of $\sqrt{n}(\widehat{\boldsymbol{\beta}}_{\texttt{QMLE}} - \boldsymbol{\beta}_0)$. Explicitly provide the "sandwich" form of the asymptotic covariance matrix $\mathbf{V}_{\texttt{QMLE}}$.

   Solution: Using Taylor expansion around $\boldsymbol{\beta}_0$: $0 = \mathbf{S}_n(\widehat{\boldsymbol{\beta}}) = \mathbf{S}_n(\boldsymbol{\beta}_0) + \mathbf{H}_n(\widetilde{\boldsymbol{\beta}})(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)$, for some value $\widetilde{\boldsymbol{\beta}}$ via the mean value theorem. Then, under regularity conditions,

$$\sqrt{n}(\widehat{\boldsymbol{\beta}}_{\texttt{QMLE}} - \boldsymbol{\beta}_0) = \boldsymbol{\Omega}_0^{-1} \frac{1}{\sqrt{n}} \mathbf{S}_n(\boldsymbol{\beta}_0) + o_{\mathbb{P}}(1),$$

   using a uniform LLN implying that $-\frac{1}{n}\mathbf{H}_n(\widetilde{\boldsymbol{\beta}}) \to_{\mathbb{P}} \boldsymbol{\Omega}_0$, and because $\frac{1}{\sqrt{n}}\mathbf{S}_n \rightsquigarrow \mathsf{Normal}(0, \boldsymbol{\Sigma}_0)$ by the CLT. Therefore,

$$\sqrt{n}(\widehat{\boldsymbol{\beta}}_{\texttt{QMLE}} - \boldsymbol{\beta}_0) \rightsquigarrow \mathsf{Normal}(0, \boldsymbol{\Omega}_0^{-1} \boldsymbol{\Sigma}_0 \boldsymbol{\Omega}_0^{-1}).$$

   The asymptotic covariance matrix is $\mathbf{V}_{\texttt{QMLE}} = \boldsymbol{\Omega}_0^{-1} \boldsymbol{\Sigma}_0 \boldsymbol{\Omega}_0^{-1}$.

4. *Poisson QMLE: Robust Standard Errors.* Given your result in the previous part, propose a consistent estimator for the asymptotic covariance matrix $\mathbf{V}_{\texttt{QMLE}}$ that is valid even if the Poisson variance assumption fails. This is often called the Huber-Eicker-White or "sandwich" estimator.

   Solution: To estimate $\mathbf{V}_{\texttt{QMLE}}$ consistently without assuming the Poisson variance, we use sample analogs $\widehat{\mathbf{\Omega}}$ and $\widehat{\mathbf{\Sigma}}$:

$$\widehat{\mathbf{\Omega}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{X}_i\mathbf{X}_i^\top \mu_i(\widehat{\boldsymbol{\beta}}_{\texttt{QMLE}}) \qquad \text{and} \qquad \widehat{\mathbf{\Sigma}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{X}_i\mathbf{X}_i^\top \big(Y_i - \mu_i(\widehat{\boldsymbol{\beta}}_{\texttt{QMLE}})\big)^2.$$

The robust estimator is $\widehat{\mathbf{V}}_{\texttt{QMLE}} = \widehat{\mathbf{\Omega}}^{-1}\widehat{\mathbf{\Sigma}}\widehat{\mathbf{\Omega}}^{-1}$.

5. *WNLS Estimator: Objective Function and FOC.* Consider the objective function:

$$Q_n(\boldsymbol{\beta}) = \sum_{i=1}^{n} w(\mathbf{X}_i)(Y_i - \exp(\mathbf{X}_i^\top\boldsymbol{\beta}))^2$$

where $w(\mathbf{X}_i) > 0$ are weights. Derive the first-order condition for $\widehat{\boldsymbol{\beta}}_{\texttt{WNLS}}$.

Solution: Differentiating with respect to $\boldsymbol{\beta}$ we have

$$\nabla_{\boldsymbol{\beta}} Q_n(\boldsymbol{\beta}) = \sum_{i=1}^{n} -2w(\mathbf{X}_i)(Y_i - \exp(\mathbf{X}_i^\top\boldsymbol{\beta}))\exp(\mathbf{X}_i^\top\boldsymbol{\beta})\mathbf{X}_i.$$

The first-order condition is

$$\sum_{i=1}^{n} w(\mathbf{X}_i)\mu_i(\boldsymbol{\beta})\mathbf{X}_i(Y_i - \mu_i(\boldsymbol{\beta})) = 0.$$

6. *WNLS Estimator: Optimal Weights.* Determine the specific weight function $w^*(\mathbf{X}_i)$ that makes the WNLS estimator asymptotically equivalent to the Poisson MLE.

Solution: The Poisson MLE score condition is $\sum_{i=1}^{n}\mathbf{X}_i(Y_i - \mu_i) = 0$. Comparing this to the WNLS FOC from above, we see the terms inside the summation match if

$$w(\mathbf{X}_i)\mu_i(\boldsymbol{\beta}) = 1 \implies w^*(\mathbf{X}_i) = \frac{1}{\mu_i(\boldsymbol{\beta})}.$$

Using weights inversely proportional to the variance (under the assumption that $Y_i|\mathbf{X}_i \sim \mathsf{Poisson}(\mu_i(\boldsymbol{\beta}_0))$) yields an efficient estimator.

7. *Misspecification.* Suppose now that the mean function is actually misspecified, i.e., $\mathbb{E}[Y_i|\mathbf{X}_i] \neq \exp(\mathbf{X}_i^\top\boldsymbol{\beta})$ for any $\boldsymbol{\beta}$. Discuss the convergence limit of the Poisson QMLE $\widehat{\boldsymbol{\beta}}_{\texttt{QMLE}}$. Does it converge to a meaningful parameter?

Solution: If the mean is misspecified, $\widehat{\boldsymbol{\beta}}_{\texttt{QMLE}} \to_{\mathbb{P}} \boldsymbol{\beta}^*$, where $\boldsymbol{\beta}^*$ minimizes the Kullback-Leibler divergence between the true density underlying the distribution of the data and the Poisson family. Specifically, $\boldsymbol{\beta}^*$ is the solution to the population score equation:

$$\mathbb{E}[\mathbf{X}_i(Y_i - \exp(\mathbf{X}_i^\top\boldsymbol{\beta}^*))] = 0.$$

This $\boldsymbol{\beta}^*$ provides the best approximation of the true conditional mean by an exponential function in a mean-squared error sense (weighted by the Poisson variance structure).

3 Uniform Consistency of the Kernel Density Estimator (50 points)

Let $X_1, \ldots, X_n$ be independent and identically distributed (i.i.d.) random variables taking values in $\mathbb{R}$ with a common probability density function f. Consider the Kernel Density Estimator:

$$\widehat{f}_n(x) = \frac{1}{nh_n} \sum_{i=1}^{n} K\Big(\frac{x - X_i}{h_n}\Big)$$

where $h_n > 0$ is the bandwidth sequence and $K : \mathbb{R} \to \mathbb{R}$ is the kernel function. We assume throughout that $h_n \to 0$ as $n \to \infty$.

We impose the following assumptions.

- The density f is bounded (i.e., $\sup_x f(x) \leq \|f\|_\infty < \infty$) and uniformly continuous on $\mathbb{R}$.
- The kernel K is supported on $[-1, 1]$, integrates to one, is symmetric about 0, is bounded (i.e., $\sup_u |K(u)| \leq \|K\|_\infty < \infty$), and is Lipschitz continuous with constant L_K (i.e., $|K(u) - K(v)| \leq L_K|u - v|$).

This question proves the uniform consistency of the kernel density estimator estimator over a compact interval D:

$$\sup_{x \in D} |\widehat{f}_n(x) - f(x)| = o_{\mathbb{P}}(1), \qquad \text{as } n \to \infty,$$

where $D = [0, 1]$ (without loss of generality), and under specific rates on h_n (to be set below). The following well-known exponential probability inequality will play a crucial role.

- **Theorem (Bernstein's Inequality)**. Let $W_1, \ldots, W_n$ be independent zero-mean random variables such that $|W_i| \leq M$ almost surely. Then,

$$\mathbb{P}\Big(\Big|\sum_{i=1}^{n} W_i\Big| > t\Big) \leq 2 \exp\left(-\frac{t^2/2}{\sum_{i=1}^{n} \mathbb{V}ar[W_i] + Mt/3}\right)$$

Please answer the following questions.

1. *Bias-Variance Decomposition.* Decompose the error $|\widehat{f}_n(x) - f(x)|$ into a bias term (deterministic) and a stochastic term (involving deviations from the mean). Explicitly define $\mathbb{E}[\widehat{f}_n(x)]$.

   Solution: We have

$$|\widehat{f}_n(x) - f(x)| \leq \underbrace{|\widehat{f}_n(x) - \mathbb{E}[\widehat{f}_n(x)]|}_{\text{Stochastic Term } Z_n(x)} + \underbrace{|\mathbb{E}[\widehat{f}_n(x)] - f(x)|}_{\text{Bias Term } B_n(x)},$$

   where $\mathbb{E}[\widehat{f}_n(x)] = \int \frac{1}{h} K(\frac{x-y}{h}) f(y) dy$.

2. *Pointwise Bias.* Show that for any fixed x, the bias converges to zero. That is, show $\lim_{n \to \infty} |\mathbb{E}[\widehat{f}_n(x)] - f(x)| = 0$.

Solution: Let $u = \frac{x-y}{h_n}$, so $y = x - h_n u$ and $dy = -h_n du$. Then, $\mathbb{E}[\widehat{f}_n(x)] = \int K(u) f(x - h_n u)du$, and hence

$$|\mathbb{E}[\widehat{f}_n(x)] - f(x)| = \left| \int K(u)(f(x - h_n u) - f(x))du \right| \le \int K(u)|f(x - h_n u) - f(x)|du.$$

Since f is continuous, $|f(x - h_n u) - f(x)| \to 0$ as $h_n \to 0$. By Dominated Convergence Theorem (bounded by $2\|f\|_\infty K(u)$), the integral goes to 0.

3. *Uniform Bias.* Show that the bias converges uniformly to zero. That is, $\lim_{n\to\infty} \sup_{x\in D} |\mathbb{E}[\widehat{f}_n(x)] - f(x)| = 0$.

   Solution: Since f is uniformly continuous, for every $\eta > 0$, there exists $\delta > 0$ such that $|y - x| < \delta \implies |f(y) - f(x)| < \eta$. If we restrict the integral to the compact support of K (where $|u| \le 1$), then $|(x - h_n u) - x| = h_n |u| \le h_n$. Choose n large enough so $h_n < \delta$. Then $\sup_x |f(x - h_n u) - f(x)| < \eta$. Then,

$$\sup_{x\in D} |B_n(x)| \le \int K(u)\eta \, du = \eta,$$

   and hence the uniform bias goes to 0.

4. *Stochastic Term.* Let $Z_n(x) = \widehat{f}_n(x) - \mathbb{E}[\widehat{f}_n(x)]$, and express this statistic as a sum of independent zero-mean random variables $Y_i(x)$. Explicitly write out the form of $Y_i(x)$.

   Solution: We have

$$Z_n(x) = \sum_{i=1}^n \left(\frac{1}{nh_n} K\left(\frac{x - X_i}{h_n}\right) - \mathbb{E}\left[\frac{1}{nh_n} K\left(\frac{x - X_i}{h_n}\right)\right] \right) = \sum_{i=1}^n Y_i(x)$$

   where $Y_i(x) = \frac{1}{nh_n}\left(K\left(\frac{x-X_i}{h_n}\right) - \mathbb{E}\left[K\left(\frac{x-X_1}{h_n}\right)\right]\right)$.

5. *Bounding the Variance and Magnitude.* To apply a concentration inequality, we need bounds on the variance and the absolute magnitude of the terms $Y_i(x)$.

   (a) Show that $|Y_i(x)| \le \frac{2\|K\|_\infty}{nh_n}$.

   (b) Show that $\mathbb{V}ar[Y_i(x)] \le \frac{\|f\|_\infty \|K\|_2^2}{n^2 h_n}$ (or a bound of order $O(\frac{1}{n^2 h_n})$).

   Solution: For part (a), note that $|K(\cdot)| \le \|K\|_\infty$. Thus, the term inside the bracket is bounded by $2\|K\|_\infty$, and we have $|Y_i(x)| \le \frac{2\|K\|_\infty}{nh_n} \equiv M_n$. For part (b), we have

$$\begin{aligned}
\mathbb{V}ar[Y_i(x)] &= \frac{1}{n^2 h_n^2} \mathbb{V}ar\left(K\left(\frac{x - X_i}{h_n}\right)\right) \le \frac{1}{n^2 h_n^2} \mathbb{E}\left[K^2\left(\frac{x - X_i}{h_n}\right)\right] \\
&= \frac{1}{n^2 h_n^2} \int K^2\left(\frac{x - y}{h_n}\right) f(y)dy = \frac{1}{n^2 h_n} \int K^2(u) f(x - h_n u)du \\
&\le \frac{\|f\|_\infty \int K^2(u)du}{n^2 h_n} = \frac{C_v}{n^2 h_n} \equiv \sigma_n^2.
\end{aligned}$$

6. *Exponential Concentration.* Apply Bernstein's inequality to $Z_n(x)$ to find an upper bound for $\mathbb{P}[|Z_n(x)| > \epsilon]$. Your bound should be of the form $2\exp(-C \cdot nh_n\epsilon^2)$ for small ϵ.

   Solution: The sum of variances is $\sum_{i=1}^n \mathbb{V}ar(Y_i) \leq n\sigma_n^2 = \frac{C_v}{nh_n}$, and the envelop term is $M = \frac{2\|K\|_\infty}{nh_n}$. Thus,

$$\mathbb{P}(|Z_n(x)| > \epsilon) \leq 2\exp\left(-\frac{\epsilon^2/2}{\frac{C_v}{nh_n} + \frac{2\|K\|_\infty \epsilon}{3nh_n}}\right) = 2\exp\left(-nh_n \frac{\epsilon^2}{2C_v + \frac{4}{3}\|K\|_\infty \epsilon}\right).$$

   For small ϵ, this behaves like $2\exp(-Cnh_n\epsilon^2)$.

7. *Discretization (Covering Argument).* Since the supremum is over an uncountable set $D = [0,1]$, we cannot simply sum probabilities. We must discretize: construct a grid $x_1, \ldots, x_{N_n}$ of points in D such that for any $x \in D$, there exists a grid point x_k with $|x - x_k| \leq \delta_n$. How many points N_n are required if we choose a spacing of δ_n?

   Solution: We cover $D = [0,1]$ with a grid of spacing δ_n. The number of points is $N_n = \lceil 1/\delta_n \rceil$.

8. *Lipschitz Continuity of the Process.* Using the Lipschitz property of the kernel K, find a bound for the difference $|Z_n(x) - Z_n(x')|$ in terms of $|x - x'|$. Specifically, show that:

$$|Z_n(x) - Z_n(x')| \leq \frac{C_L}{h_n^2}|x - x'|$$

   for some constant C_L independent of x, n, h_n.

   Solution: Using $|K(u) - K(v)| \leq L_K|u - v|$,

$$\begin{aligned} |\widehat{f}_n(x) - \widehat{f}_n(x')| &\leq \frac{1}{nh_n}\sum \left| K\left(\frac{x - X_i}{h_n}\right) - K\left(\frac{x' - X_i}{h_n}\right)\right| \\ &\leq \frac{1}{nh_n}\sum L_K \left|\frac{x - x'}{h_n}\right| = \frac{L_K}{h_n^2}|x - x'| \end{aligned}$$

   The same Lipschitz constant applies to $\mathbb{E}[\widehat{f}_n(x)]$. Thus, $|Z_n(x) - Z_n(x')| \leq \frac{2L_K}{h_n^2}|x - x'|$.

9. *The Union Bound.* Show that

$$\sup_{x \in D} |Z_n(x)| \leq \max_{k=1,\ldots,N_n} |Z_n(x_k)| + \sup_{x,x':|x-x'|<\delta_n} |Z_n(x) - Z_n(x')|,$$

   and choose δ_n (as a function of h_n, n, ϵ) such that the second term (the discretization error) is strictly less than $\epsilon/2$. Apply the Union Bound and the Bernstein result (Part 6) to the first term (the grid maximum).

Solution: We need the discretization error $\frac{2L_K}{h_n^2}\delta_n \leq \frac{\epsilon}{2}$. Set $\delta_n = \frac{\epsilon h_n^2}{4L_K}$. Then, $N_n \approx \frac{4L_K}{\epsilon h_n^2}$.

$$\mathbb{P}\left(\sup_{x \in D} |Z_n(x)| > \epsilon\right) \leq \mathbb{P}\left(\max_{1 \leq k \leq N_n} |Z_n(x_k)| > \epsilon/2\right) \leq N_n \mathbb{P}(|Z_n(x_1)| > \epsilon/2)$$
$$\leq \frac{C'}{\epsilon h_n^2} \exp(-C'' n h_n \epsilon^2).$$

10. *Conclusion.* Determine the sufficient condition on the rate of decay of h_n for uniform consistency, that is, $\sup_{x \in D} |\widehat{f}_n(x) - f(x)| = o_{\mathbb{P}}(1)$.

    Solution: For the probability to go to 0, we need the exponent to dominate the pre-factor. That is,

    $$\exp(-C'' n h_n \epsilon^2) \cdot h_n^{-2} \to 0.$$

    Taking logs: $-C'' n h_n \epsilon^2 - 2\log(h_n) \to -\infty$. This essentially requires $nh_n \to \infty$ fast enough to kill $\log(1/h_n)$. More formally, if we assume $h_n = n^{-\alpha}$, we generally require $\frac{\log n}{nh_n} \to 0$. If this holds, uniform consistency is achieved.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m$, $x = a_1, ..., a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m$, $t \in \mathcal{R}$
$DU(a_1, ..., a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m$, $\bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}$, $m = 1, 2, ...$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, 1, ..., n$
	m.g.f.	$(pe^t + 1 - p)^n$, $t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1]$, $n = 1, 2, ...$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!$, $x = 0, 1, 2, ...$
	m.g.f.	$e^{\theta(e^t - 1)}$, $t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p$, $x = 1, 2, ...$
	m.g.f.	$pe^t/[1 - (1-p)e^t]$, $t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-	p.d.f.	$\binom{n}{x}\binom{m}{r-x} / \binom{N}{r}$
geometric		$x = 0, 1, ..., \min\{r, n\}$, $r - x \le m$
	m.g.f.	No explicit form
$HG(r, n, m)$	Expectation	rn/N
	Variance	$rnm(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, ...$, $N = n + m$
Negative	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$, $x = r, r+1, ...$
binomial	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r$, $t < -\log(1-p)$
	Expectation	r/p
$NB(p, r)$	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1]$, $r = 1, 2, ...$
Log-	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x$, $x = 1, 2, ...$
distribution	m.g.f.	$\log[1 - (1-p)e^t]/\log p$, $t \in \mathcal{R}$
	Expectation	$-(1-p)/(p \log p)$
$L(p)$	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt}-e^{at})/[(b-a)t],\ \ t\in\mathcal{R}$
$U(a,b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a,\ b\in\mathcal{R},\ \ a<b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t+\sigma^2t^2/2},\ \ t\in\mathcal{R}$
$N(\mu,\sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1},\ \ t<\theta^{-1}$
$E(a,\theta)$	Expectation	$\theta+a$
	Variance	θ^2
	Parameter	$\theta>0,\ \ a\in\mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2},\ \ t<1/2$
χ^2_k	Expectation	k
	Variance	$2k$
	Parameter	$k=1,2,...$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha},\ \ t<\gamma^{-1}$
$\Gamma(\alpha,\gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma>0,\ \alpha>0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha,\beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha>0,\ \ \beta>0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t-\sigma\lvert t\rvert}$
$C(\mu,\sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu\in\mathcal{R},\ \ \sigma>0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)}\left(1+\frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	$0, \ (n>1)$
	Variance	$n/(n-2), \ (n>2)$
	Parameter	$n=1,2,...$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2), \ (m>2)$
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, $(m>4)$
	Parameter	$n=1,2,..., \ m=1,2,...$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x-\mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu,\sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2}-1)$
	Parameter	$\mu\in\mathcal{R}, \ \sigma>0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha,\theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1}+1)$
	Variance	$\theta^{2/\alpha}\left\{\Gamma(2\alpha^{-1}+1)-\left[\Gamma(\alpha^{-1}+1)\right]^2\right\}$
	Parameter	$\theta>0, \ \alpha>0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{-\lvert x-\mu\rvert/\theta}$
	m.g.f.	$e^{\mu t}/(1-\theta^2t^2), \ \lvert t\rvert<\theta^{-1}$
	Expectation	μ
$DE(\mu,\theta)$	Variance	$2\theta^2$
	Parameter	$\mu\in\mathcal{R}, \ \theta>0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a,\theta)$	Expectation	$\theta a/(\theta-1), \ (\theta>1)$
	Variance	$\theta a^2/[(\theta-1)^2(\theta-2)], \ (\theta>2)$
	Parameter	$\theta>0, \ a>0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1+e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1+\sigma t)\Gamma(1-\sigma t), \ \lvert t\rvert<\sigma^{-1}$
$LG(\mu,\sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu\in\mathcal{R}, \ \sigma>0$